

Name: _____



Using the fx-991CW for number and algebra

Task A: Conversions

1) Convert the following times to hours.

- a) 1 hour and 15 mins
- b) 2 hours, 20 mins and 20 secs
- c) 4 hours, 15 mins and 16 secs

2) Convert the following number of hours to hours, minutes and seconds.

- a) 7.25
- b) 3.7
- c) 7.42
- d) 3.63

3) Fill in the gaps.

- a) $121 = \square^2$
- b) $5500 = 2\square \times \square^3 \times 11$
- c) $384 = \square\square \times \square\square$
- d) $351 = 3\square \times \square\square \times 13$

Task B: Solving linear equations

1) Solve the following without using a calculator and then check your answers using the calculator.

Which way of solving is faster?

$$\text{a) } \frac{1}{2}x + 7 = 25 \quad \text{b) } \frac{3x - 5}{2} = \frac{2x + 1}{10} \quad \text{c) } 8(y + 5) = 13 - \frac{y + 5}{2}$$

2) Solve the following pairs of simultaneous equations using your calculator.

$$\begin{aligned} \text{a) } 2a + 3b &= 1 \\ -a + 4b &= 8 \end{aligned}$$

$$\begin{aligned} \text{b) } 2b + a &= 8 \\ 2b + 3 &= 7a \end{aligned}$$



Task C: Solving polynomials

1) Solve these quadratic inequalities (to 2 d.p. if appropriate).

a) $-x^2 - 8x + 20 \geq 0$

b) $x^2 + 8x - 19 < 0$

c) $x^2 + 6x + 8 < 3$

d) $5x^2 - x + 5 > 3x^2 - 10x - 2$